

Job Sequencing and Tool Switching

Structural results and a branch-and-Benders-cut study

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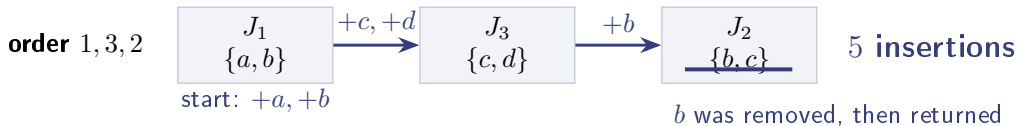
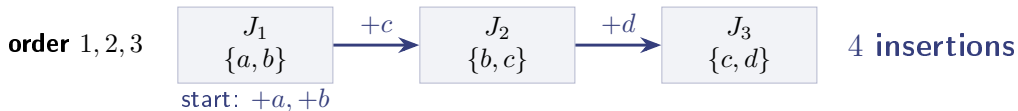
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Why the order matters

Magazine capacity $b = 2$; the machine starts empty.



Fixed order

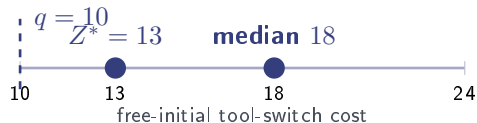
When a tool must leave, remove the loaded tool whose next use is furthest away.

$$\text{KTNS}(\pi) \text{ in } O(n|U|).$$

Choosing the order is NP-hard for every fixed $b \geq 2$.

(Tang & Denardo, 1988; Crama et al., 1994)

40,320
orders of the eight-job instance L1-1
180
are optimal



Where the problem arises

CNC cells
tool magazine

PCB assembly
feeder slots

Colour printing
ink cartridges

Order picking
what a picker carries

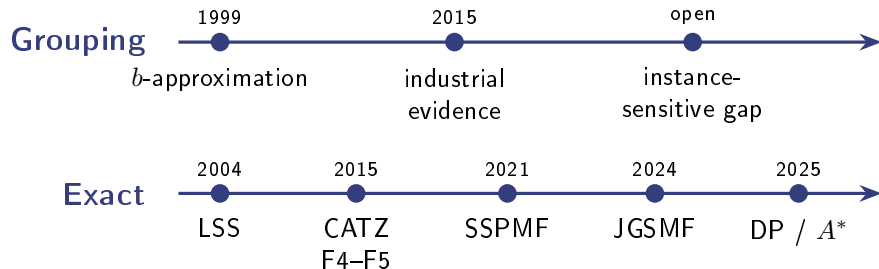
Same combinatorial core

A capacity- b resource, jobs that need subsets of it, and a changeover cost paid between consecutive jobs.

Why the order matters

A cell running thousands of parts a day repeats the changeover constantly, so the sequence is chosen once and paid for every shift.

(Burger et al., 2015)



Two things the timeline does not contain: a bound on the two-phase loss that depends on the **instance**, and a relaxation that gets **past the counting bound**.

(Crama & van de Klundert, 1999; Catanzaro et al., 2015; da Silva et al., 2021)

Question 1 – structural

Industry does not solve the SSP. It groups jobs into as few magazine-fitting batches as possible, then sequences the batches.

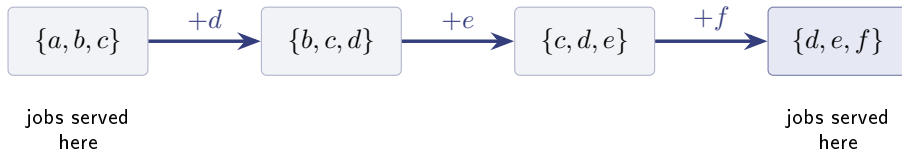
What does insisting on the fewest groups cost, once the final KTNS step is allowed to re-optimize the loading?

Question 2 – algorithmic

Every compact model surveyed relaxes to the counting bound q , so it is fast where q is exact and adrift where it is not.

Can a method that never forms that relaxation – an exact loading oracle, or a reformulation over configurations – get past it?

Part I answers the first and finds an unbounded loss. **Part II** answers the second, and the answer is no — for a reason worth naming.



A schedule is a walk

Magazine configurations are the nodes; the step cost is $d(C, C') = |C' \setminus C|$, the tools that must be inserted. Every job must be served at some node that contains its requirement.

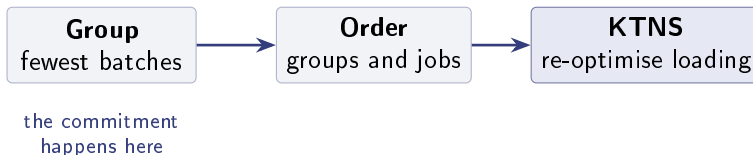
Hence one counting bound

Each used tool enters at least once, and at most b of them can already be loaded at the start:

$$Z^* \geq q = |U| - b.$$

This single quantity governs **both** halves of the talk.

The two-phase heuristic

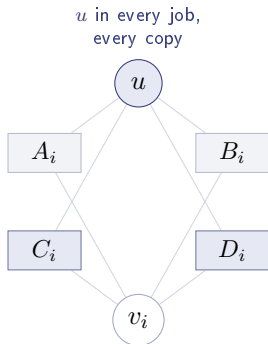


Where the loss comes from. Allowing *all* feasible groupings makes the construction **exact**: the optimum equals the cheapest walk over every grouping. So the loss is not a defect of the method — it is the price of insisting on the **fewest** groups.

Two readings must be separated: the walk through one fixed magazine per group, and the value after KTNS re-optimises. **They differ.**

(Crama & van de Klundert, 1999)

The gap is unbounded on connected instances



Theorem

For every $g \geq 1$ there is a **connected** instance with $b = 6$,

$$|U| = 8g+1, \quad K^* = 3g, \quad Z^* = 8g-5, \quad H = 9g-5.$$

Hence $H - Z^* = g$: the loss of the two-phase method is **unbounded**, even on connected instances.

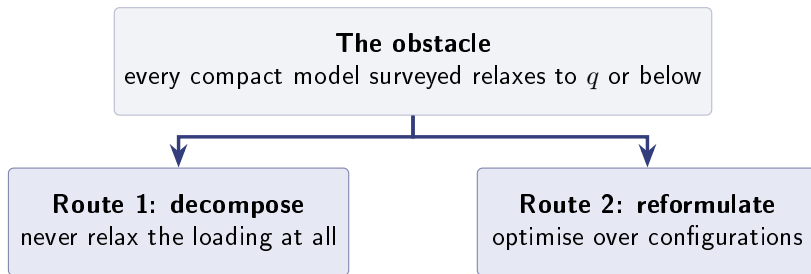
Take g copies of a four-job seed, add one universal tool and a private marker per copy. The minimum grouping becomes *unique*, and each copy is forced to let one private tool leave and return.

The **ratio** tends to $9/8$: the additive loss grows, the factor does not.

Result	Status
Optimum = cheapest walk over all groupings	proved
Loss = price of minimum cardinality (walk value)	proved
Zero-gap classes for every capacity	proved
Ratio bound $\min(b, K^* - 1, q)$, uniform over tie-breaking	proved
Closed form and extremal behaviour at three groups	proved
Four jobs necessary and sufficient for a positive loss	proved
Connected family with unbounded additive loss	proved
Constant-factor bound at $b = 3$	open

In the finite censuses, positive gaps are rare and none occurs at $b = 2$.

Two ways past the coverage bound



Branch-and-Benders-cut

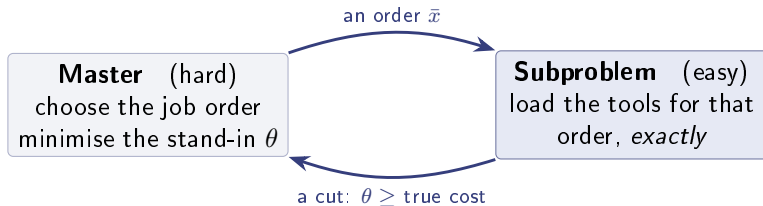
The loading subproblem is polynomial, so it becomes an exact oracle and the master never forms a weak relaxation *of the loading*.

Position-indexed branch-and-price

Columns are magazine configurations. One variant prices *transitions*, and its relaxation can provably exceed q .

(da Silva et al., 2021; Rahmaniani et al., 2017)

A branch-and-Benders-cut algorithm



Why this problem invites Benders. For a **fixed** order the loading is solvable in polynomial time. So the subproblem is not relaxed — it is solved exactly, and the master's bound is built from true subproblem values rather than from a weak relaxation of the loading.

(Hooker & Ottosson, 2003; Codato & Fischetti, 2006; Rahmaniani et al., 2017)

The master problem

Arc variables $x_{ij} \in \{0, 1\}$ on jobs plus a depot d marking start and end, so the x describe a Hamiltonian path.

$$\begin{aligned} \min \theta \quad \text{s.t.} \quad & \sum_{i \neq j} x_{ij} = 1 \quad (\forall j), & \sum_{j \neq i} x_{ij} = 1 \quad (\forall i), \\ & \sum_{i,j \in S} x_{ij} \leq |S| - 1 \quad (S \subseteq J), & \theta \geq |U|. \end{aligned}$$

Two bounds the master starts from. The **coverage row** $\theta \geq |U|$, and a **pairwise row** $\theta \geq \sum_{i,j} w_{ij} x_{ij} + \sum_j |T_j| x_{dj}$ with $w_{ij} = \max(0, |T_i \cup T_j| - b)$.

The subproblem: loading for a fixed order

Fix the order $\bar{x}$. Let $y_{j,t} = 1$ if tool t is in the magazine while job j runs, and $z_{j,t}$ charge an insertion of t at j .

$$\begin{aligned} \min \quad & \sum_{j,t} z_{j,t} \\ & y_{j,t} \geq 1 && t \in T_j \quad (\nu_{jt}) \\ & \sum_t y_{j,t} \leq b && \forall j \quad (\mu_j) \\ & z_{j,t} \geq y_{j,t} - y_{i,t} - (1 - \bar{x}_{ij}) && \forall i \neq j, t \quad (\lambda_{ijt}) \\ & z_{j,t} \geq y_{j,t} - (1 - \bar{x}_{dj}) && \forall j, t \quad (\lambda_{jt}^d) \end{aligned}$$

Exactness of the loading oracle

For fixed $\bar{x}$ the master variables sit only on the right-hand side. The remaining matrix has the consecutive-ones property along the selected path and is totally unimodular, so the relaxation is **integral** and its optimum is exactly the KTNS cost of that order.

(Tang & Denardo, 1988)

Only $\bar{x}$ appears on the right-hand side, so in the dual it appears only in the objective. Taking an optimal dual $(\bar{\lambda}, \bar{\lambda}^d, \bar{\mu}, \bar{\nu})$:

$$\theta \geq \sum_{i \neq j} \sum_t \bar{\lambda}_{ijt} (x_{ij} - 1) + \sum_j \sum_t \bar{\lambda}_{jt}^d (x_{dj} - 1) - b \sum_j \bar{\mu}_j + \sum_j \sum_{t \in T_j} \bar{\nu}_{jt}$$

Valid globally. The dual feasible region does not depend on $\bar{x}$, so the cut holds for every order.

Tight locally. At $\bar{x}$ it reproduces the exact loading cost. It is affine in x , so fractional separation needs no separate argument.

A defect found during the campaign. The depot sum $\sum_{j,t} \bar{\lambda}_{jt}^d (x_{dj} - 1)$ was initially omitted. The cut then became **too tight**: it cut off genuine optima.

How it surfaced

Randomly generated instances were solved by brute force and by the solver. **193** violations of the true optimum were recorded.

What was done

The term was restored, the derivation was rewritten from the implementation, and the entire campaign was re-run.

The first magazine is charged in full because the machine starts empty — dropping it silently assumes the first job's tools are free.

LP dual cuts (BBC-LP)

The cut above, from an optimal dual of the loading LP at an integer master solution.

Combinatorial cuts (BBC-K)

A no-good style cut built directly from the KTNS value of the proposed order.

Four strengthenings, each a standard remedy.

- +F **fractional separation** — cut at fractional master points, not only integer ones
- +T **triplet bounds** — a three-job strengthening of the pairwise row
- +C **conflict-graph row** — jobs that cannot share a magazine
- +P **Pareto-optimal lifting** — choose the deepest among alternative optimal duals
- +H **hybrid incumbent** — a genetic heuristic run at the root

The reasoning

When a Benders scheme tails off, the standard fix is to separate at *fractional* points rather than waiting for integer ones. The cut is affine in x , so this is legitimate.

It did what it should, locally

67.5 million cuts generated. Median node count fell from **77,439** to **3,701** — a factor of twenty.

And the bound never moved. Across 1,404 paired runs, the dual bound at termination was

higher	0
equal	1,226
lower	178

and **69** instances were *lost*.

The remedy is applied correctly, to the wrong obstacle.

Configuration	What it adds	Solved
BBC-LP	—	724
BBC-LP+T	triplet bounds	723
BBC-LP+F+H	hybrid incumbent	690
BBC-LP+F	fractional separation	655
BBC-LP+F+C	conflict-graph row	654
BBC-LP+F+P	Pareto lifting	618

The pattern. Only the strengthening that improves the **incumbent** helps (+31 over its own reference). Every strengthening aimed at the **dual bound** is neutral or harmful. A solver whose difficulty was *finding* good solutions would not behave this way.

Compact baselines

SSPMF

CATZ-F4

LSS

Reimplementations used for a controlled comparison, not a literature-wide ranking.

Benders regimes

BBC-LP

BBC-LP+T

BBC-K

BBC-LP+F

BBC-K+F

BBC-LP+F+H

BBC-LP+F+C

BBC-LP+F+P

BBC-LP+ACC

1,410 canonical benchmark instances

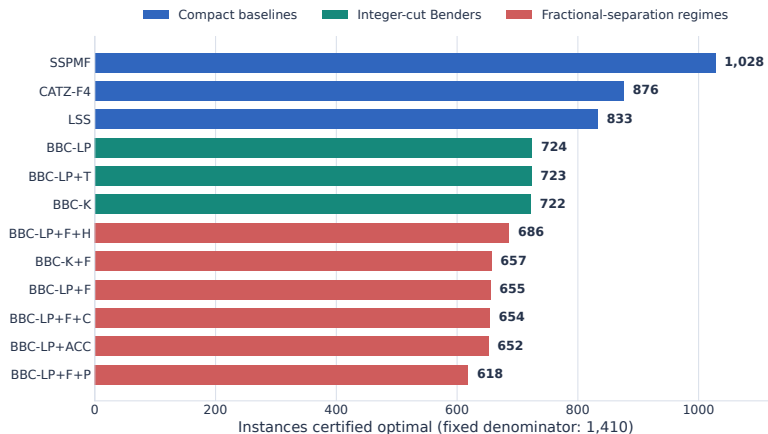
12 solver configurations

16,920 canonical outcomes

1 h per run, fixed denominator

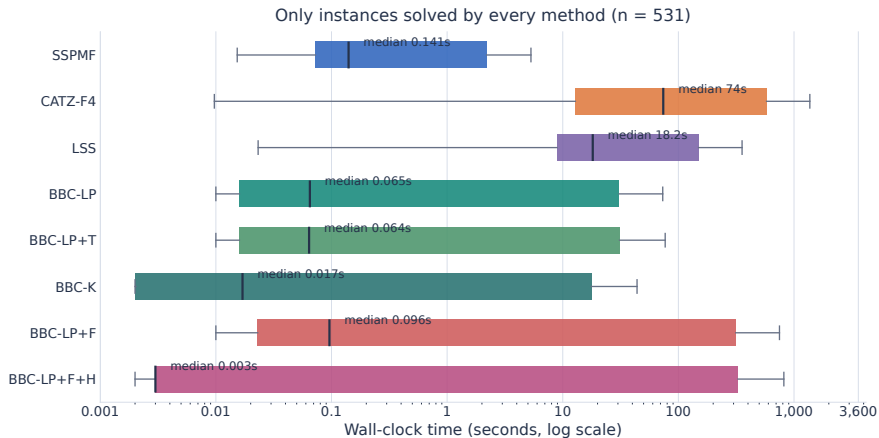
(Laporte et al., 2004; Catanzaro et al., 2015; da Silva et al., 2021)

How far each method gets



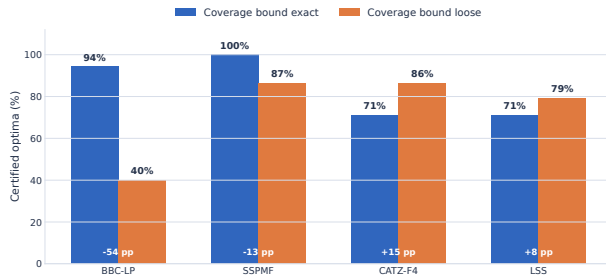
Each bar uses the same 1,410-instance denominator.

Solving times

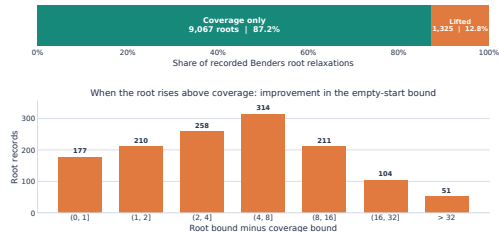


The same 531 instances appear in every box; the horizontal axis is logarithmic.

The coverage bound and what each method closes



1,107 instances with a known optimum.



223/383 time-limited,
known-optimum
BBC-LP incumbents already equal
the optimum.

What the theory isolates

Minimum grouping can force a tool out and back:

$$\text{cost} = q + \text{reinsertions}.$$

What the campaign isolates

The tested master often finds the right order but cannot certify those returns.

The work narrows the next experiment; it claims no new SOTA solver.

Next exact tests

- disaggregate θ by position;
- encode job-union structure over a window;
- test the stronger bound under a matched protocol.

Use the grouping result

- search $K^*, \dots, K^* + \delta$ groupings;
- flatten each with KTNS; use the orders as seeds.

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- Hooker, J. N., and Ottosson, G. (2003). Logic-based Benders decomposition. *Mathematical Programming*, vol. 96, pp. 33–60.
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- Rahmaniani, R., Crainic, T. G., Gendreau, M., and Rei, W. (2017). The Benders decomposition algorithm: a literature review. *European Journal of Operational Research*, vol. 259, no. 3, pp. 801–817.
- Tang, C. S., and Denardo, E. V. (1988). Models arising from a flexible manufacturing machine, Part I: minimization of the number of tool switches. *Operations Research*, vol. 36, no. 5, pp. 767–777.

Every work listed here is cited on a slide of this talk. The full bibliography is in the report.

Thank You!

Questions

The obstacle. Every compact model surveyed relaxes to the counting bound $q = |U| - b$ or below. A branch-and-cut built on any of them is fast where q is exact and adrift where it is not.

Reformulating over **magazine configurations** attacks this from the other side. A configuration is a set $C \subseteq U$ with $|C| = b$; these become the columns, generated on demand. Three formulations were built.

	Column	Linear relaxation
PCF	configuration at a position	$= q$ with the counting rows, 0 without
PCF'	prefix form of PCF	$= q$, symmetry removed
PTF	<i>transition</i> between two	$\geq q$, and can exceed it

$y_C^k = 1$ if configuration C occupies position k ; $w_t^k \geq 0$ counts an insertion of t at position k .

$$\begin{aligned} \min \quad & \sum_{t,k} w_t^k \\ \text{(P)} \quad & \sum_C y_C^k = 1, & k = 1, \dots, n \\ \text{(Cov)} \quad & \sum_k \sum_{C \supseteq T_j} y_C^k \geq 1, & j \in J \\ \text{(W)} \quad & w_t^k \geq \sum_{C \ni t} (y_C^k - y_C^{k-1}), & t \in T, k \geq 2 \\ \text{(T)} \quad & \sum_{k \geq 2} w_t^k + \sum_{C \ni t} y_C^1 \geq 1, & t \in U \end{aligned}$$

Proposition

Without (T) the relaxation is **zero** — one convex blend of configurations, repeated at every position, satisfies (P), (Cov) and (W) at no cost. With (T) it is exactly q . *Proved, and confirmed on all 2,108 recorded roots.*

Backup: the position–transition formulation

$z_{C,C'}^k \geq 0$ is a column for the step from position k to $k + 1$ leaving C and entering C' , at cost $d(C, C') = |C' \setminus C|$.

$$\min \sum_k \sum_{(C,C')} d(C, C') z_{C,C'}^k$$

$$(S) \quad \sum_{(C,C')} z_{C,C'}^k = 1, \quad k = 1, \dots, n - 1$$

$$(F) \quad \sum_{(C,C'): t \in C'} z_{C,C'}^k = \sum_{(D,D'): t \in D} z_{D,D'}^{k+1}, \quad t \in T$$

(Cov) every job served at the head or tail of some step

$$(T) \quad \sum_k \sum_{(C,C'): t \notin C, t \in C'} z_{C,C'}^k + \sum_{(C,C'): t \in C} z_{C,C'}^1 \geq 1, \quad t \in U$$

Proposition

A column *carries a step*, so the relaxation cannot mix configurations without paying the transition. Hence $\text{LP(PTF)} \geq q$, and it can be strictly greater — the only construction in this work that provably exceeds q .

The pricing problem

Choosing the entering configuration is

NP-hard: pick b tools to collect per-tool duals plus a bonus for each job fully covered. PTF prices *coupled pairs*, so a node costs far more again.

Rows stay small

$O(n|T|)$ and fixed; only the columns are generated.

What was measured.

- PCF' root = q on **all** 2,108 recorded runs — the proposition confirmed on the benchmark.
- PTF exceeds q on **2 of 232** roots, rising from 5 to 6 where the optima are 12 and 13.
- Nothing above nine jobs was closed: a Python pricer against compiled solvers.

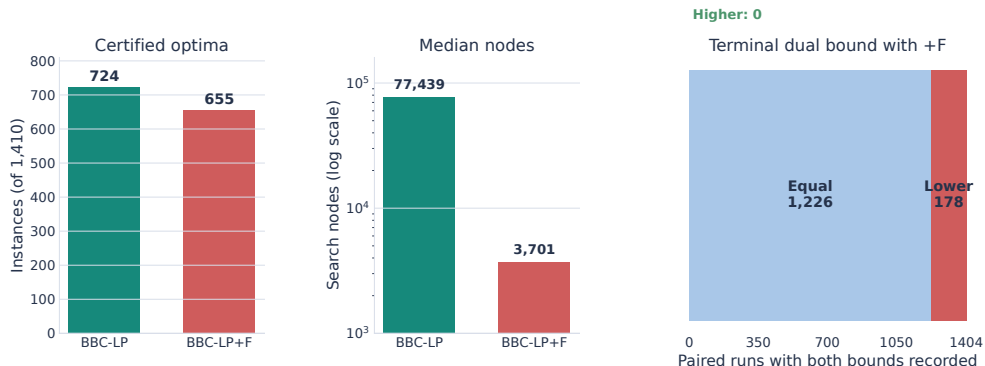
Sound, and as it stands too rare and too small to close the certificate gap.

Backup: results by instance family

Configuration	Cat.	Crama	Lap.3	Lap.4	Lap.5	Lap.7	All
SSPMF	62	68	340	330	162	66	1028
CATZ-F4	44	44	340	330	74	44	876
LSS	41	42	340	330	39	41	833
BBC-LP+T	34	34	340	192	84	39	723
BBC-LP	34	35	340	192	84	39	724
BBC-K	35	35	340	190	83	39	722
BBC-LP+F+H	39	37	316	157	97	40	686
BBC-LP+F	33	32	313	156	85	36	655
BBC-LP+F+C	33	32	311	157	85	36	654
BBC-LP+ACC	39	37	286	154	97	39	652
BBC-K+F	33	31	316	156	86	35	657
BBC-LP+F+P	31	32	282	155	83	35	618

Family denominators: 160, 160, 340, 330, 340, 80; total 1,410.

Backup: the fractional-cut ablation



BBC-LP+F generated 67.0 million fractional cuts. The +F regime also disables presolve.

Against BBC-LP+F: $+H$ certifies 31 more; $+C$ certifies 1 fewer; $+P$ certifies 37 fewer; all three together certify 3 fewer.

Connected-family proof

$K^* = 3g$	unique feasible pair in each copy
$Z^* = 8g - 5$	coverage bound plus a schedule with no return
$H = 9g - 5$	one forced return per copy plus a matching order
connected	universal tool u lies in every job

Independent checks

349,872	loading checks against an exact dynamic program
988	instances proved optimal by two methods
41,514	pairwise method comparisons, none disagreeing
16,920	canonical campaign outcomes audited

Formal scope and remaining conjectures: THEOREM_AUDIT.md.